

Benchmark Investigation of Panda: A Pretrained Forecast Model for Chaotic Dynamics

Experiment Log and Analysis

UNSW Global Taste of Research India Summer Practicum 2026

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June 2026

Abstract

This document records all benchmark experiments conducted on Panda (Lai, Bao, Gilpin, ICLR 2026), a pretrained forecast model for chaotic dynamical systems. The investigation was motivated by Prof. Salim’s direction to test Panda’s emergent capabilities on standard time series and spatiotemporal benchmarks. Each experiment is reported with its motivation, methodology, methodological limitations, direct observations, and competing explanations. Claims are explicitly labelled as: **[OBS]** observed in data, **[PAT]** empirical pattern across multiple conditions, **[HYP]** proposed explanation, **[SPEC]** speculative, or **[EST]** established by prior theory or paper. This labelling convention was decided to maintain distinction between observation and possible interpretations rather than mixing them.

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1 Experimental Setup

1.1 Models

Two models are compared throughout all experiments.

Panda 21M. The base checkpoint from `GilpinLab/panda` on HuggingFace. A 21 million parameter encoder-only transformer trained exclusively on 20,000 synthetic chaotic ordinary differential equations (ODEs). Prediction horizon 128 steps. Context length 512 steps. For horizons exceeding 128 steps, window autoregression is used: predictions are fed back as context for subsequent rollouts.

Chronos 20M. The `amazon/chronos-t5-small` checkpoint. A 20 million parameter causal univariate decoder model trained on a large corpus of real-world time series. Chronos forecasts each channel independently.

The two models are matched in parameter count, making comparisons attributable to architecture and training distribution rather than scale.

1.2 Evaluation Protocol

Unless stated otherwise, the following protocol is used across all experiments.

Normalisation. Each context window is normalised independently using its own mean and standard deviation per channel (per-window instance normalisation). This prevents global statistics from leaking future information into each evaluation window.

Evaluation windows. A sliding window of context length 512 is moved across the time series. The number of windows is specified per experiment. In early experiments (pre-presentation), $n_{\text{windows}} = 6\text{--}10$ was used. In the fixed experiment set, $n_{\text{windows}} = 8$ or $n_{\text{windows}} = 20$ is used as noted.

Primary metric. Mean Absolute Error (MAE) is the primary metric throughout. Symmetric Mean Absolute Percentage Error (sMAPE) was used in earlier experiments but was found to be unreliable on instance-normalised data where values near zero cause the denominator to vanish, artificially inflating or deflating the metric.

Statistical test. Where stated, the Wilcoxon signed-rank test (one-sided, alternative: model A better than model B) is applied to the per-window MAE values. With $n_{\text{windows}} = 8$, the minimum achievable p-value is 0.004. With $n_{\text{windows}} = 20$, the minimum achievable p-value is approximately 0.0001. All p-values should be interpreted in light of these floors.

Advantage. Panda advantage is defined as $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$. Positive values indicate Panda is better.

1.3 Hardware

All experiments were run on a CPU-only laptop. Runtimes are consequently long for experiments involving many channels or long horizons.

2 Pre-Presentation Experiments

These experiments were conducted prior to the group meeting and form the basis of the initial findings presented to Prof. Salim.

2.1 Experiment 1: Standard Time Series Benchmarks (Initial)

2.1.1 Motivation

To establish whether Panda has any advantage on non-chaotic standard benchmarks. ETTh1, ETTh2, and Weather were chosen as the canonical time series forecasting benchmarks used by TimesNet, PatchTST, and Chronos comparisons in the literature.

2.1.2 Method

Datasets: ETTh1 (7 channels, 17,420 timesteps), ETTh2 (7 channels, 17,420 timesteps), Weather (21 channels, 52,696 timesteps). Prediction horizon: 128 steps. $n_{\text{windows}} = 10$. Primary metric: sMAPE. Instance normalisation applied globally to the full dataset before windowing (not per-window).

2.1.3 Methodological Limitations

1. sMAPE is unreliable on instance-normalised data where channel values are near zero. Results are not quantitatively trustworthy.
2. Global normalisation leaks future statistics into evaluation windows.
3. Single prediction horizon (128 steps). Results may not generalise to other horizons.
4. No statistical significance tests.
5. $n_{\text{windows}} = 10$ provides limited statistical power.

2.1.4 Observations

[OBS] Both models produced sMAPE approximately 0.8–0.9 on all three datasets. **[OBS]** Panda slightly worse than Chronos on ETTh1. **[OBS]** Chronos sMAPE = 0.009 on Lorenz rho = 15 was flagged as a metric artifact rather than genuine model performance.

2.1.5 Explanation Shipped at the Time

The conclusion presented was: “Panda fails on non-chaotic data because it was trained on stationary chaotic ODEs with no seasonal inductive bias.” This was stated as a finding despite the invalid metric and the limitation to a single prediction horizon.

2.1.6 Alternative Explanations Not Presented

1. **[HYP]** The result may be specific to $H = 128$. Later fixed experiments showed Panda wins on Weather at all tested horizons and on ETTh1 and ETTh2 at longer horizons.
2. **[HYP]** The sMAPE artifact may have masked the true comparison entirely, making the result uninterpretable.
3. **[HYP]** Weather and ETT differ substantially in channel count and coupling structure. Grouping them as “non-chaotic data” ignored this distinction.

Confidence: Very Low. The original conclusion is largely contradicted by the fixed Experiment 8.

2.2 Experiment 2: Double Pendulum Graded Noise

2.2.1 Motivation

To quantify the cost of measurement noise on a chaotic system and test whether Panda’s advantage over Chronos is robust to noise. This also served as a proxy for the gap between clean synthetic data and noisy real experimental data.

2.2.2 Method

The double pendulum ODE was simulated with Gaussian observation noise at six levels: $\sigma \in \{0, 0.01, 0.05, 0.10, 0.25, 0.50\}$ times the signal standard deviation. One random seed, one initial condition. $n_{\text{windows}} = 8$. sMAPE and MAE both reported.

2.2.3 Methodological Limitations

1. Single seed and single initial condition. One realisation of the trajectory.
2. The real experimental dataset (Asseman et al. 2018) was not obtained. The experiment was described as “synthetic versus real” but was in practice “synthetic versus synthetic with Gaussian noise.” Gaussian additive noise is not the same type of nonstationarity as real experimental measurement error.
3. Noise levels were not tested beyond $\sigma = 0.50$.
4. No statistical significance tests.

2.2.4 Observations

[OBS] Panda MAE at $\sigma = 0$: 0.62. Panda MAE stays approximately flat (0.63–0.67) across all tested noise levels up to $\sigma = 0.50$. **[OBS]** Chronos MAE at $\sigma = 0$: 0.71. Chronos MAE peaks at approximately 0.77 at high noise. **[OBS]** Panda consistently lower MAE than Chronos at every tested noise level. **[OBS]** sMAPE panel showed erratic non-monotone behaviour, correctly identified as a metric artifact on normalised data.

2.2.5 Explanation Shipped at the Time

“The random Fourier and polynomial features are smooth functions of the input, making the embedding noise-robust. Takens’ theorem is noise-tolerant.” This was stated as a conclusion rather than a hypothesis.

2.2.6 Alternative Explanations Not Presented

1. **[HYP]** Chronos uses a discrete tokenisation scheme. Continuous Gaussian noise may specifically disrupt this tokenisation, independently of any property of Panda’s architecture.
2. **[HYP]** Panda’s predictions may have lower variance by default. A flat MAE noise curve could reflect low-variance predictions rather than genuine noise robustness.
3. **[HYP]** The tested noise levels may be below the breakdown threshold for both models. The claim cannot be extrapolated to higher noise regimes.

Confidence: Medium. for the observation. **Confidence: Low.** for the proposed mechanism.

2.3 Experiment 3: Lorenz Rho Sweep

2.3.1 Motivation

To test whether Panda’s advantage specifically tracks the onset of chaos. The Lorenz parameter ρ controls the transition from periodic ($\rho < 24.74$) to chaotic ($\rho \geq 24.74$) dynamics. This provides a controlled test of the chaos-threshold hypothesis.

2.3.2 Method

$\rho \in \{10, 15, 20, 24, 24.74, 26, 28, 35, 45, 60\}$. One trajectory per ρ value, one initial condition. $n_{\text{windows}} = 8$. MAE as primary metric. Maximum Lyapunov exponent estimated using the Rosenstein method. The estimator required three implementation attempts before producing physically reasonable output.

2.3.3 Methodological Limitations

1. The Lyapunov estimator systematically overestimates. For the Lorenz system at $\rho = 28$, the literature value is $\lambda_1 \approx 0.9$ but the estimator returned 1.66. The x-axis of Lyapunov-based figures is not quantitatively reliable.
2. Single trajectory and single seed per ρ value.
3. $n_{\text{windows}} = 8$ with no confidence intervals on advantage values.
4. Results at $\rho = 10, 15$ showed Chronos sMAPE near zero in the periodic regime, correctly identified as an artifact.

2.3.4 Observations

ρ	Regime	$\hat{\lambda}_1$	Panda MAE	Chronos MAE	Advantage
10.00	Periodic	-0.29	≈ 0.02	≈ 0.02	≈ 0
15.00	Periodic	-0.33	≈ 0.04	≈ 0.04	≈ 0
20.00	Near transition	0.20	≈ 0.09	≈ 0.31	+0.22
24.00	Near transition	0.33	≈ 0.10	≈ 0.44	+0.34
24.74	Onset of chaos	1.75	≈ 0.18	≈ 0.65	+0.47
28.00	Chaotic	1.66	≈ 0.25	≈ 0.89	+0.64
60.00	Chaotic	1.63	≈ 0.48	≈ 1.17	+0.69

Table 1: Lorenz ρ sweep results (approximate values from $n_{\text{windows}} = 8$, one seed).

[OBS] Estimated λ_1 is negative at $\rho = 10, 15$ and jumps to 1.75 at $\rho = 24.74$, consistent with the theoretical bifurcation point. **[OBS]** Panda MAE advantage is approximately zero at $\rho = 10, 15, 20$. The advantage jumps near $\rho = 24.74$ and remains positive across all tested chaotic ρ values. **[OBS]** No tested chaotic ρ value produced a result where Chronos had lower MAE than Panda. **[PAT]** The pattern of advantage increasing with ρ is consistent across the chaotic range, though not strictly monotone.

2.3.5 *Explanation Shipped at the Time*

“Panda’s advantage is specifically activated by the chaotic transition. The positive Lyapunov exponent is necessary and sufficient for Panda’s advantage.” This was stated as a conclusion.

2.3.6 *Alternative Explanations Not Presented*

1. **[HYP]** Signal statistics (variance, spectral content, amplitude) change discontinuously at the bifurcation. Any model better suited to high-variance aperiodic signals would show this pattern, independently of any chaos-specific mechanism. This alternative is not yet ruled out.
2. **[HYP]** Chronos tokenisation may fail specifically on aperiodic continuous signals, making Chronos worse in the chaotic regime regardless of Panda’s capabilities.
3. **[HYP]** The advantage may be driven primarily by the Koopman dynamics embedding rather than any chaos-specific learning. This is not testable without an ablation removing the dynamics embedding.

What would falsify the claim: A chaotic system where Panda does not win, or a non-chaotic system where Panda wins substantially. The Weather and Burgers sweep results (Experiments 8 and 10) partially address this.

Confidence: Medium. for the observation. **Confidence: Low.** for the chaos-specific mechanism hypothesis.

2.4 Experiment 4: `dysts` Systems and Advantage versus `Lambda1`

2.4.1 *Motivation*

To test Panda’s advantage across a broader set of known chaotic systems from the `dysts` library and to measure whether a Spearman correlation between estimated λ_1 and Panda advantage reaches statistical significance.

2.4.2 *Method*

Systems tested: Lorenz, Rossler, Chua, Duffing, Halvorsen, SprottB, Thomas, Rucklidge, Dadras, Bouali, DequanLi. $n_{\text{windows}} = 8$ per system. Lyapunov estimates via Rosenstein method. Additional systems were added iteratively until the Spearman p-value crossed 0.05.

2.4.3 *Methodological Limitations*

1. **P-hacking.** Systems were added iteratively until $p < 0.05$ was achieved. The reported Spearman $\rho = 0.370$, $p = 0.044$ is not a valid pre-registered result. The true p-value corrected for sequential testing is unknown but higher.
2. Systems in the `dysts` library are not independent (several share structural similarities).
3. Lyapunov estimates are noisy and systematically biased (see Experiment 3).
4. Single seed per system. No cross-system multiple comparison correction.

2.4.4 Observations

[OBS] Panda wins (lower MAE) on every tested `dysts` chaotic system. [OBS] Spearman $\rho = 0.370$, $p = 0.044$ after iterative addition of systems. This p-value is not reliable due to p-hacking. [OBS] Two visual sub-clusters: Lorenz family above the trend line, double pendulum below the trend line.

2.4.5 Explanation Shipped at the Time

“Panda has learned something universal about chaotic dynamics. Chaos is necessary and sufficient. In-distribution systems show larger advantage.” Overstated given the methodological problems.

2.4.6 Alternative Explanations Not Presented

1. [HYP] The correlation may be driven entirely by the Lorenz family cluster. Removing those points may eliminate significance entirely.
2. [HYP] The sub-cluster structure (Lorenz family above trend, double pendulum below) suggests training-distribution effects rather than universality. This directly challenges the “universal chaos learning” claim.

Confidence: Low. due to p-hacking. The observation that Panda wins on every tested chaotic system is more robust (**Confidence: Medium.**) but still limited to $n = 8$ windows per system.

2.5 Experiment 5: Burgers Viscosity Sweep (First Version)

2.5.1 Motivation

To test whether the chaos-threshold pattern found in the Lorenz rho sweep extends to partial differential equations (PDEs). Burgers equation viscosity ν controls the transition from smooth diffusion-dominated to chaotic shock-dominated dynamics, analogous to ρ in Lorenz.

2.5.2 Method

$\nu \in \{0.1, 0.05, 0.02, 0.01, 0.005\}$. PCA reduction to 16 channels. $n_{\text{windows}} = 8$. Lyapunov estimated from first PCA component. Solver diverged at $\nu = 1.0$, leaving no non-chaotic baseline.

2.5.3 Methodological Limitations

1. No non-chaotic baseline. Without $\nu \geq 0.5$, it is impossible to verify a threshold effect.
2. Lyapunov estimation returned approximately zero for all conditions. The chaos-threshold connection could not be verified from the data.
3. Non-monotone reversal of advantage at $\nu = 0.02 \rightarrow 0.01$ was not discussed.
4. PCA representation changes character as ν changes, confounding the comparison across conditions.

2.5.4 Observations

[OBS] Panda advantage positive at all tested ν values. **[OBS]** Advantage roughly increases with decreasing ν but with a non-monotone reversal at $\nu = 0.02 \rightarrow 0.01$.

Current status: Superseded by the fixed Experiment 10, which includes a non-chaotic baseline and a stable solver.

2.6 Experiment 6: PCA versus Spatial Subsampling (First Version)

2.6.1 Motivation

To compare two methods for converting a 1D spatiotemporal PDE field into channels for Panda: global PCA modes versus raw spatial location subsampling.

2.6.2 Method

Single ν value (0.005). Single spatial location ($x = 0$) for subsampling. $n_{\text{windows}} = 6$.

2.6.3 Critical Flaw

The spatial location $x = 0$ was a near-nodal point of the Burgers field with near-constant dynamics. Both models trivially predicted near-zero values and scored well. No general conclusion about spatial subsampling is possible from one degenerate location. This experiment was rerun in fixed Experiment 12 with four methods including a variance-stratified control.

2.6.4 Observations

[OBS] Both models performed approximately $8\times$ better in absolute MAE on spatial subsampling than on PCA. **[OBS]** Panda relative advantage larger with PCA (0.093) than subsampling (0.044).

Confidence: Very Low. for any conclusion. Single degenerate location.

3 Fixed Experiments

These experiments address the methodological problems identified in the pre-presentation set. A new notebook (`fixed_experiments.ipynb`) was created. Key improvements: per-window normalisation, increased n_{windows} , Wilcoxon signed-rank tests, multiple prediction horizons, non-chaotic baselines, and removal of oracle leakage.

3.1 Experiment 7: Standard Horizon Evaluation (Week 1, Invalid)

The Week 1 version of standard horizon evaluation retained global normalisation leakage and used $n_{\text{windows}} = 6$ with sMAPE as primary metric. Results were superseded by Experiment 8 and are not reported here.

3.2 Experiment 8: Standard Horizon Evaluation (Fixed)

3.2.1 Motivation

To establish the true performance gap between Panda and Chronos on standard non-chaotic benchmarks across the full range of prediction horizons used in the literature.

3.2.2 Method

Datasets: ETTh1 (7 channels), ETTh2 (7 channels), Weather (21 channels). Horizons: $H \in \{96, 192, 336, 720\}$. $n_{\text{windows}} = 20$. Per-window normalisation. Wilcoxon signed-rank tests. Weather $H = 720$ was not completed due to hardware time constraints.

3.2.3 Observations

Dataset	H	Panda MAE	IQR	Chronos MAE	IQR	Adv (p)
ETTh1	96	0.7269	0.2203	0.6633	0.2388	−0.064 (0.844)
ETTh1	192	0.8185	0.3046	0.7825	0.2742	−0.036 (0.826)
ETTh1	336	0.8571	0.1971	0.9013	0.1859	+0.044 (0.649)
ETTh1	720	0.9921	0.3226	1.0189	0.3845	+0.027 (0.774)
ETTh2	96	0.8736	0.4790	0.9494	0.5583	+0.076 (0.478)
ETTh2	192	0.9697	0.5380	0.9505	0.5730	−0.019 (0.147)
ETTh2	336	0.9255	0.3860	1.1101	0.3831	+0.185 (0.013)*
ETTh2	720	1.1139	0.2194	1.1027	0.3862	−0.011 (0.392)
Weather	96	0.6378	0.1723	0.8115	0.2036	+0.174 (0.000)*
Weather	192	0.7224	0.2385	0.9582	0.2089	+0.236 (0.001)*
Weather	336	0.8481	0.2898	1.0843	0.3507	+0.236 (0.000)*

Table 2: Fixed Experiment 8. * denotes $p < 0.05$ (Wilcoxon, $n = 20$ windows).

[OBS] On ETTh1, no horizon shows statistically significant difference in either direction. All p-values exceed 0.6. The direction of advantage is inconsistent across horizons. **[OBS]** On ETTh2, one horizon ($H = 336$) shows significant Panda advantage ($p = 0.013$). Other horizons are not significant and inconsistent in direction. **[OBS]** On Weather, Panda has statistically significant advantage at $H = 96, 192, 336$ (all $p \leq 0.001$). Advantage magnitude is 0.17–0.24, consistent across horizons. **[PAT]** Weather is the only dataset where Panda shows consistent, statistically significant advantage across multiple horizons.

3.2.4 Competing Explanations for the Weather Result

1. **[HYP]** Weather has 21 channels versus 7 for ETT. More channels provide more information regardless of coupling. Any multivariate model might benefit.
2. **[HYP]** Weather has genuine multivariate physical coupling (pressure drives wind drives temperature). Channel attention may capture this.
3. **[HYP]** Chronos is specifically poorly suited to Weather’s complex multi-scale spectral structure (10-minute sampling, 144-step daily cycle).
4. **[SPEC]** Panda handles quasi-chaotic components of Weather (atmospheric turbulence at synoptic scales) better than Chronos due to its chaotic training distribution.

Confidence: High. for the Weather observation ($n = 20$, $p < 0.001$). **Confidence: Low.** for any proposed mechanism.

3.3 Experiment 9: Univariate Ablation on Weather

3.3.1 Motivation

To directly test whether channel attention drives Panda’s advantage on Weather. If channel attention is responsible, removing it (by forecasting each channel independently) should substantially reduce Panda’s advantage over Chronos.

3.3.2 Method

Multivariate Panda (all 21 channels processed jointly) was compared against univariate Panda (each channel processed independently, suppressing cross-channel attention) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

3.3.3 Observations

H	panda_uni	panda_multi	Adv (uni better)	p
96	0.5541	0.6113	+0.057	0.074
336	0.8467	0.8762	+0.030	0.371

Table 3: Univariate ablation. Positive advantage means univariate Panda has lower MAE than multivariate Panda.

[OBS] At both horizons, univariate Panda has lower MAE than multivariate Panda. **[OBS]** Neither result is statistically significant at $\alpha = 0.05$ ($p = 0.074$, $p = 0.371$). **[OBS]** Effect size is small (0.03–0.06 MAE difference).

3.3.4 What This Establishes and Does Not Establish

The data is inconsistent with the hypothesis that channel attention drives Panda’s Weather advantage. Channel attention does not help and may marginally hurt on Weather. However the effect is small and not significant at $n = 8$ windows. A definitive conclusion requires more windows.

3.3.5 Competing Explanations

1. **[HYP]** The temporal architecture (non-causal encoder, Koopman patch embedding, fixed prediction head) is sufficient to beat Chronos. Channel attention adds noise on non-chaotic data. **Confidence: Medium.:** consistent with data.
2. **[HYP]** $n = 8$ is too small to detect a real but small positive channel attention effect. The true effect may be positive but undetectable. **Confidence: Medium.:** cannot rule out.
3. **[SPEC]** Channel attention trained on chaotic ODE coupling learns patterns counterproductive on periodic non-chaotic data. **Confidence: Low.:** speculative.

Confidence: Medium. for the observation. The channel attention hypothesis is not supported but not conclusively falsified.

3.4 Experiment 10: Burgers Viscosity Sweep (Fixed)

3.4.1 Motivation

To test the chaos-threshold hypothesis for PDEs with a proper non-chaotic baseline. The stable solver prevents divergence at high ν , enabling comparison across the full smooth-to-chaotic range.

3.4.2 Method

$\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.02, 0.01, 0.005\}$. Adaptive timestep solver (CFL stability condition satisfied at all ν). PCA to 16 channels. $n_{\text{windows}} = 8$. $H = 128$.

3.4.3 Observations

ν	Physical regime	Panda MAE	Chronos MAE	Adv (p)
2.000	Diffusion dominated	0.0152	0.0191	+0.004 (0.273)
1.000	Diffusion dominated	0.0186	0.0568	+0.038 (0.004)*
0.500	Transition	0.0402	0.1023	+0.062 (0.004)*
0.100	Weakly nonlinear	0.1127	0.2239	+0.111 (0.004)*
0.050	Shock forming	0.1374	0.2865	+0.149 (0.004)*
0.020	Strong shocks	0.1394	0.2593	+0.120 (0.004)*
0.010	Strong shocks	0.1375	0.2325	+0.095 (0.004)*
0.005	Strong shocks	0.1509	0.2732	+0.122 (0.004)*

Table 4: Fixed Burgers viscosity sweep. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows.

[OBS] At $\nu = 2.0$, Panda advantage is +0.004, $p = 0.273$. Not significant. This is the strongly diffusion-dominated regime. **[OBS]** At $\nu = 1.0$, $p = 0.004$. Significant. Burgers at $\nu = 1.0$ is diffusion-dominated and non-chaotic by standard criteria ($\lambda_1 \leq 0$). **[OBS]** Panda wins significantly at $\nu = 1.0$ and $\nu = 0.5$, which are not chaotic regimes. **[OBS]** Advantage is not monotonically increasing with decreasing ν . It peaks near $\nu = 0.05$ (0.149), drops at $\nu = 0.02$ (0.120) and $\nu = 0.01$ (0.095), then rises at $\nu = 0.005$ (0.122).

3.4.4 Critical Finding

[PAT] The chaos-threshold pattern observed in the Lorenz ρ sweep does not replicate cleanly in the Burgers viscosity sweep. Panda wins significantly at non-chaotic viscosity values ($\nu = 1.0$, $\nu = 0.5$) where a chaos-specific mechanism would predict no advantage.

3.4.5 Competing Explanations

1. **[HYP]** Even at $\nu = 1.0$, the PCA channels of the Burgers field are spatially coupled through the diffusion operator. Panda’s channel attention may capture coupling that is present even in non-chaotic smooth flows. **Confidence: Medium..**
2. **[HYP]** Chronos is poorly suited to PCA modal time series regardless of chaoticity. The advantage reflects Chronos weakness rather than Panda strength in the non-chaotic regime. **Confidence: Medium.:** consistent with decomposition results.
3. **[HYP]** The single non-significant result at $\nu = 2.0$ versus the significant result at $\nu = 1.0$ may be noise. The true threshold may lie between 1.0 and 2.0 rather than at the expected chaos onset. **Confidence: Medium.:** cannot rule out with $n = 8$.
4. **[SPEC]** Burgers PCA modes may have spectral characteristics that look dynamically complex to Panda even at non-chaotic ν , due to the mode-mixing nature of PCA on a spatially heterogeneous field.

What would distinguish these: Estimating λ_1 from the PCA component time series at each ν value using the corrected Rosenstein estimator. If $\lambda_1 < 0$ at $\nu = 1.0$ but Panda still wins significantly, explanations 1, 2, or 3 are supported over a chaos-specific mechanism.

Confidence: Medium. for the observation. **Confidence: Low.** for any specific mechanism.

3.5 Experiment 11: FFT Decomposition Preprocessing (Fixed)

3.5.1 Motivation

To test whether removing periodic and trend components from a time series before presenting it to Panda changes Panda’s relative advantage. An earlier version of this experiment (Week 1) contained oracle leakage: the future trend and seasonal components were extracted from the full series and given directly to both models. The fixed version uses only context-window statistics.

3.5.2 Method

FFT decomposition applied to each channel of the context window only. Deterministic component (trend plus seasonal harmonics) projected forward naively: linear trend extrapolation plus repetition of the last full seasonal cycle. Both models forecast the residual; the projected deterministic is added back before evaluation. Comparison against vanilla (no decomposition) run in the same evaluation loop. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

3.5.3 Note on the Invalid Earlier Version

The oracle-leaking version of this experiment reported a 65% reduction in MAE for both models after decomposition. The corrected version shows a 25% increase in Panda MAE on ETTh1 at $H = 336$. The two results are in opposite directions. This confirms that

the earlier result was entirely an artifact of providing future periodic information to both models.

3.5.4 Observations

Dataset	Cond.	H	Panda	Chronos	Adv (p)
ETTh1	Vanilla	96	0.7205	0.8157	+0.095 (0.422)
ETTh1	Decomp	96	0.8032	0.8293	+0.026 (0.578)
ETTh1	Vanilla	336	0.8407	0.8856	+0.045 (0.273)
ETTh1	Decomp	336	1.0474	1.0174	−0.030 (0.770)
ETTh2	Vanilla	96	0.9329	1.0409	+0.108 (0.230)
ETTh2	Decomp	96	0.8856	0.9640	+0.078 (0.273)
Weather	Vanilla	96	0.6082	0.6880	+0.080 (0.020)*
Weather	Decomp	96	0.6771	0.6871	+0.010 (0.230)
Weather	Vanilla	336	0.8762	0.9987	+0.114 (0.004)*
Weather	Decomp	336	1.0609	1.0760	+0.015 (0.020)*

Table 5: Fixed FFT decomposition experiment. * denotes $p < 0.05$.

[OBS] Decomposition increases absolute MAE for both models on ETTh1 at $H = 336$ (Panda +25%, Chronos +15%) and on Weather at $H = 96$ (Panda +11%, Chronos +0.1%). **[OBS]** On Weather, decomposition collapses Panda’s advantage. Vanilla $H = 96$: advantage = +0.080, $p = 0.020$. Decomp $H = 96$: advantage = +0.010, $p = 0.230$. **[OBS]** At $H = 336$ on Weather, the decomp advantage remains statistically significant ($p = 0.020$) but the effect size is very small (+0.015 versus vanilla +0.114).

3.5.5 Competing Explanations

1. **[HYP]** Panda’s advantage on Weather comes from better handling of the periodic and trend components. On the residual, both models are comparable. **Confidence: Medium.**: consistent with all available data.
2. **[HYP]** The naive FFT projection introduces large errors in the projected deterministic component, degrading both models. The result reflects projection error rather than genuine comparison on the residual. **Confidence: Medium.**: the 25% MAE increase on ETTh1 is suspicious and may reflect projection error.
3. **[SPEC]** The periodic component of Weather contains genuine multivariate coupling. Removing it leaves a weakly coupled residual where channel attention is irrelevant.

Confidence: High. for the observation. **Confidence: Low.** for any mechanism.

3.6 Experiment 12: Subsampling Methods (Fixed)

3.6.1 Motivation

To test whether dynamical diversity subsampling of spatial locations gives Panda a larger advantage than uniform or variance-stratified subsampling. The variance-stratified baseline was introduced to control for the confound in the earlier experiment, where a degenerate near-zero-variance location artificially improved both models.

3.6.2 Method

Four methods: (1) Uniform spacing, (2) Variance-stratified uniform (excludes bottom 10% variance locations before uniform spacing), (3) PCA (global spatial modes), (4) Diversity (farthest-point sampling in a dynamical feature space: standard deviation, mean absolute value, 90th percentile amplitude, spectral entropy). $\nu \in \{0.05, 0.005\}$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 128$.

3.6.3 Observations

ν	Method	Panda MAE	Chronos MAE	Adv	p
0.05	Uniform	0.0317	0.0765	+0.045	0.004
0.05	Stratified	0.0296	0.0736	+0.044	0.004
0.05	PCA	0.1374	0.2915	+0.154	0.004
0.05	Diversity	0.0317	0.1132	+0.082	0.004
0.005	Uniform	0.0288	0.1696	+0.141	0.004
0.005	Stratified	0.0301	0.1779	+0.148	0.004
0.005	PCA	0.1509	0.3029	+0.152	0.004
0.005	Diversity	0.0300	0.2475	+0.218	0.004

Table 6: Fixed subsampling comparison. All results $p = 0.004$ (minimum achievable with $n = 8$ windows).

[OBS] All results reach $p = 0.004$, the minimum achievable Wilcoxon p-value with $n = 8$ windows. The p-values are therefore uninformative for comparing methods against each other. **[OBS]** Panda absolute MAE across Uniform, Stratified, and Diversity methods is nearly identical at both ν values. At $\nu = 0.005$: Uniform = 0.0288, Stratified = 0.0301, Diversity = 0.0300. **[OBS]** Chronos MAE varies substantially across methods. Diversity locations produce higher Chronos MAE (0.2475) than Uniform locations (0.1696) at $\nu = 0.005$. **[OBS]** Diversity advantage (0.218) exceeds Stratified advantage (0.148) at $\nu = 0.005$. However, the difference is entirely attributable to Chronos MAE (0.2475 vs 1.779). Panda MAE is the same (0.0300 vs 0.0301).

3.6.4 Critical Observation

[PAT] Panda’s absolute forecasting performance is invariant to the spatial subsampling method (Uniform, Stratified, Diversity). Diversity subsampling increases Panda’s relative advantage over Chronos by making the problem harder for Chronos, not easier for Panda.

3.6.5 Competing Explanations for the Diversity Effect

1. **[HYP]** Diversity locations are more dynamically informative. Chronos cannot handle them but Panda can, demonstrating a genuine advantage of Panda’s architecture. **Confidence: Low.:** speculative, not supported by Panda’s flat absolute MAE.
2. **[HYP]** Diversity sampling selects spatially extreme or high-variance locations that break Chronos pattern matching without providing additional information to Panda. **Confidence: Medium.:** consistent with the observation that Panda MAE is unchanged.

3. **[HYP]** PCA gives the largest relative advantage because global modes are harder for Chronos (high complexity, global structure) while Panda handles them adequately due to the Koopman embedding. **Confidence: Medium..**

Confidence: Medium. for the invariance of Panda absolute MAE observation. **Confidence: Low.** for any proposed mechanism.

4 Mechanistic Investigation Experiments

These experiments were designed to address the open scientific questions identified after the fixed experiment set. The primary questions were: (1) what drives Panda’s Weather advantage, (2) whether signal statistics confound the Lorenz result, (3) whether $\lambda_1 < 0$ at non-chaotic Burgers viscosities, (4) whether diversity subsampling conclusions are seed-stable, and (5) whether improved period projection changes the decomposition result.

4.1 Experiment 13: Periodic Component Forecasting on Weather (Constructed Target)

4.1.1 Motivation

To test whether Panda’s Weather advantage is specifically located in the periodic/deterministic component of the signal. If Panda’s advantage amplifies when both models are given only the periodic component as input and evaluated against a periodic target, this suggests Panda handles periodicity better than Chronos.

4.1.2 Method

FFT extraction of the top-5 frequency components (excluding DC) from each context window channel. The resulting periodic context is presented to both models. The target is constructed as the FFT sinusoidal extrapolation of those 5 harmonics into the prediction window — not the real future values of the series. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.1.3 Critical Methodological Note

The target in this experiment is a constructed FFT projection, not the real future values. Both models are therefore evaluated against a deterministic sinusoidal extrapolation rather than ground truth. This conflates “which model better reproduces FFT sinusoidal extrapolation” with “which model better forecasts the actual periodic component of the future.” This confound was identified after the run and led directly to Experiment 18 (Option A), which corrects it.

4.1.4 Observations

Condition	H	Panda advantage	p
Full signal (reference)	96	+0.1893	0.004
Periodic only (constructed target)	96	+0.5220	0.004
Full signal (reference)	336	+0.1031	0.020
Periodic only (constructed target)	336	+0.5014	0.004

Table 7: Experiment 13. Periodic-only advantage computed against FFT-constructed target, not real future values. Result superseded by Experiment 18.

[OBS] Panda advantage is approximately $2.8\times$ larger on the periodic-only condition than the full signal at $H = 96$, and $4.9\times$ larger at $H = 336$. **[OBS]** Both conditions are statistically significant ($p = 0.004$ and $p = 0.020$).

Status: Confounded by the constructed target. Superseded by Experiment 18. The amplified advantage is not interpretable in isolation.

Confidence: Very Low. for any conclusion about Panda’s periodic handling ability from this experiment alone.

4.2 Experiment 14: Lorenz Phase Surrogate Control

4.2.1 Motivation

To test whether Panda’s advantage on chaotic Lorenz ($\rho = 28$) is specific to chaotic dynamical structure, or whether it is driven by signal statistics alone (variance, power spectrum). A phase-randomization surrogate matches the power spectrum of the original signal exactly while destroying all phase relationships and temporal ordering.

4.2.2 Method

Phase-randomization surrogate applied to the Lorenz $\rho = 28$ trajectory: FFT magnitudes are preserved; phases are replaced with uniform random draws on $[0, 2\pi]$; inverse FFT is applied. The surrogate is rescaled to exactly match the original mean and standard deviation. Both models are run on the original Lorenz and on the surrogate. $n_{\text{windows}} = 8$. $H = 96$.

4.2.3 Methodological Limitations

1. $n_{\text{windows}} = 8$ provides low statistical power. Minimum achievable p-value is 0.004.
2. Phase shuffling introduces endpoint discontinuities and artificial stationarity not present in the original signal. These properties differ from the original for reasons beyond merely removing dynamical structure.
3. The log proposed matching variance and spectral entropy to the $\rho = 15$ periodic baseline. The implemented control is a phase surrogate of $\rho = 28$ itself. These are related but distinct controls.

4.2.4 Observations

Condition	Panda advantage	p
Lorenz $\rho = 28$ (chaotic)	+0.3835	0.004
Phase-shuffled surrogate	+0.1715	0.320

Table 8: Experiment 14. Phase surrogate control for Lorenz $\rho = 28$. $n_{\text{windows}} = 8$.

[OBS] Panda advantage drops by approximately 55% on the surrogate ($0.3835 \rightarrow 0.1715$).

[OBS] Panda advantage on the surrogate is not statistically significant ($p = 0.320$).

[OBS] The residual advantage of 0.1715 is not negligible in absolute terms.

4.2.5 Competing Explanations

1. [HYP] Panda’s advantage is specific to chaotic dynamical structure. Phase shuffling destroys temporal ordering and the advantage collapses. **Confidence: Medium.:** direction is consistent, but see alternative 2.

2. **[HYP]** The result is a Type II error. With $n = 8$ windows, the test has low power to detect a true advantage of 0.17. The surrogate advantage may be real but undetectable at this sample size. **Confidence: Medium.**: cannot rule out without larger n .
3. **[HYP]** Phase shuffling introduces artificial stationarity that differentially affects both models, independently of any dynamics-specific mechanism. **Confidence: Low.**: known issue with phase surrogates, magnitude unknown.

What would strengthen this: Rerun with $n_{\text{windows}} = 20$. If the surrogate advantage remains near 0.17 and still non-significant, the dynamics-specific hypothesis is strengthened.

Confidence: Medium. for the directional observation. **Confidence: Low.** for the conclusion that dynamics are the specific driver.

4.3 Experiment 15: Burgers λ_1 Estimation from PCA Components

4.3.1 Motivation

Experiment 10 showed Panda wins significantly at $\nu = 1.0$ (diffusion-dominated, non-chaotic by physical criteria). To directly verify whether $\lambda_1 < 0$ at these viscosities, the corrected Rosenstein estimator was applied to the first PCA component time series at each ν value.

4.3.2 Method

Burgers equation simulated at $\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.01, 0.005\}$ using the stable adaptive-timestep spectral solver. PCA reduction to 16 channels. Rosenstein estimator applied to first PCA component with embedding dimension $m = 3$, delay $\tau = 1$, Theiler exclusion window $w = \max(0.02N, 10)$. $n_{\text{windows}} = 8$. $H = 96$.

4.3.3 Critical Methodological Note on the Estimator

Two independent reasons render the λ_1 estimates unreliable for this system:

1. **Insufficient embedding dimension.** Rosenstein requires $m \geq 2d + 1$ where d is the attractor dimension. For Burgers at low ν , $d \sim \mathcal{O}(1/\nu)$. At $\nu = 0.005$, d may be $\mathcal{O}(100)$. With $m = 3$, nearest-neighbour search is performed in a 3-dimensional projection of a potentially 100-dimensional attractor. All points appear near all others in this projection, producing a flat divergence curve regardless of the true λ_1 .
2. **PCA modes are not generic Takens observables.** Takens' theorem guarantees attractor reconstruction only for a generic scalar observable. PCA modes are constructed to maximise explained variance, which means they capture the dominant stable spatial structure while discarding the high-wavenumber chaotic directions. The first PCA component may genuinely not observe the chaotic dynamics even when they are present.

These issues are not fixable by adjusting estimator parameters without changing the observable. All λ_1 values reported here should be treated as unreliable. Physical regime labels derived from the known Burgers bifurcation structure should be used as the chaos proxy instead.

4.3.4 Observations

ν	Physical regime	$\hat{\lambda}_1$ (unreliable)	Panda advantage	p
2.000	Diffusion dominated	NaN	−0.0015	0.191
1.000	Diffusion dominated	−0.011	+0.0045	0.004*
0.500	Transition	−0.007	+0.0156	0.012*
0.100	Weakly nonlinear	−0.004	+0.1010	0.004*
0.050	Shock forming	−0.003	+0.1187	0.004*
0.010	Strong shocks	−0.001	+0.1510	0.004*
0.005	Strong shocks	−0.001	+0.1003	0.004*

Table 9: Experiment 15. $\hat{\lambda}_1$ values are unreliable due to insufficient embedding dimension and non-generic PCA observable. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows. $H = 96$ throughout (versus $H = 128$ in Experiment 10).

[OBS] $\hat{\lambda}_1$ is negative or NaN at every tested ν , including the most shock-dominated regimes. Values cluster in the range $[-0.011, -0.001]$ with near-zero absolute magnitude across all ν . This is inconsistent with physical expectations and confirms estimator failure. **[OBS]** Panda advantage increases monotonically with decreasing ν from $\nu = 1.0$ to $\nu = 0.01$ (+0.0045 to +0.1510). A non-monotone reversal occurs at $\nu = 0.005$ (+0.1003 < +0.1510). **[OBS]** At $\nu = 2.0$, Panda advantage is −0.0015 ($p = 0.191$, not significant). Consistent with Experiment 10 at this regime. **[OBS]** Advantage values at $\nu = 1.0$ and $\nu = 0.5$ are substantially smaller here (+0.0045, +0.0156) than in Experiment 10 (+0.038, +0.062). This is attributable to the different prediction horizon ($H = 96$ here versus $H = 128$ in Experiment 10), not a contradiction.

4.3.5 Competing Explanations for the Monotonic ν –Advantage Relationship

1. **[HYP]** Lower ν produces more dynamically complex Burgers fields. Panda’s Koopman embedding captures this complexity better than Chronos regardless of whether the field is technically chaotic by λ_1 criteria. **Confidence: Medium.:** consistent with data but mechanism not isolated.
2. **[HYP]** Lower ν produces PCA modes with higher frequency content and sharper transitions. Chronos tokenisation degrades on these modes independently of any chaos-specific factor. **Confidence: Medium.:** consistent with Chronos-weakness pattern seen in Experiment 12.
3. **[HYP]** The ν sweep is confounded by changing PCA explained variance ratios. At different ν , the first PCA mode captures a different fraction of total variance, changing the effective forecast difficulty independently of dynamics. **Confidence: Medium.:** PC1 explained variance was 0.819 at $\nu = 2.0$; values at other ν were not recorded.

Confidence: Low. for any conclusion involving $\hat{\lambda}_1$. The estimator is inappropriate for PCA modal time series of high-dimensional PDEs.

4.4 Experiment 16: Multi-Seed Subsampling Variance

4.4.1 Motivation

Experiment 12 reported diversity subsampling results from a single random seed. The critical observation—that Panda absolute MAE is invariant to subsampling method—was based on one realisation. This experiment tests whether that observation is stable across seeds and whether diversity consistently outperforms stratified subsampling.

4.4.2 Method

Diversity subsampling (farthest-point sampling in dynamical feature space) and Variance-Stratified Uniform subsampling run at 10 seeds (0–9) on Burgers $\nu = 0.05$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 96$.

4.4.3 Observations

Seed	Diversity adv	Stratified adv	p (div)	p (strat)
0	+0.0677	+0.0424	0.004	0.004
1	+0.0277	+0.0455	0.004	0.004
2	+0.0849	+0.0707	0.004	0.004
3	+0.1018	+0.0362	0.004	0.004
4	+0.0310	+0.1071	0.004	0.004
5	+0.1101	+0.0610	0.004	0.004
6	+0.0716	+0.0432	0.004	0.008
7	+0.1146	+0.0729	0.004	0.004
8	+0.0826	+0.0528	0.004	0.004
9	+0.0797	+0.0324	0.004	0.004
Mean	+0.0772	+0.0564		
Std	± 0.0296	± 0.0220		

Table 10: Experiment 16. Advantage per seed for Diversity and Stratified Uniform subsampling. Burgers $\nu = 0.05$, $N_{\text{channels}} = 16$, $H = 96$.

[OBS] Panda advantage is positive at every seed and both methods (all $p \leq 0.008$). The fact of Panda winning is robust to seed choice. **[OBS]** The ordering of diversity versus stratified advantage is inconsistent across seeds. At seed 4, stratified advantage (0.1071) exceeds diversity advantage (0.0310). At seed 7, diversity advantage (0.1146) exceeds stratified (0.0729). No consistent dominance. **[OBS]** Diversity advantage standard deviation (0.0296) is 38% of its mean (0.0772). Stratified advantage standard deviation (0.0220) is 39% of its mean (0.0564). Both methods have high relative variance across seeds.

4.4.4 Critical Observation

[PAT] The single-seed finding from Experiment 12 that diversity subsampling gives larger advantage than stratified is not reliable. Across 10 seeds the two methods are not consistently ordered. The seed variance is large relative to the mean advantage for both methods.

4.4.5 Competing Explanations for High Seed Variance

1. **[HYP]** Different seeds select genuinely different dynamical regimes from the Burgers field, producing legitimately different forecast difficulties. **Confidence: Medium.:** plausible for diversity method, less so for stratified which is deterministic.
2. **[HYP]** $n_{\text{windows}} = 8$ is insufficient to average out window-to-window variance. Seed variance and window-sampling variance are confounded. **Confidence: Medium.:** cannot be separated at this sample size.
3. **[HYP]** The diversity metric (dynamical features plus farthest-point sampling) is sensitive to its random initialisation, producing substantially different channel subsets across seeds. **Confidence: Low.:** not directly verified.

Confidence: Medium. for the seed-instability observation. The single-seed Experiment 12 conclusion about diversity subsampling superiority is not supported.

4.5 Experiment 17: Improved Period Projection in Decomposition

4.5.1 Motivation

Experiment 11 used a naive projection method (repeat last full seasonal cycle) for the deterministic component. One competing explanation for the collapse of Panda’s advantage after decomposition was that the naive projection introduces large errors, degrading both models. This experiment tests a better projection: average seasonal pattern over all complete periods in the context window, with phase-aligned tiling. No future information is used.

4.5.2 Method

FFT decomposition as in Experiment 11. For the deterministic component projection, instead of repeating the last period, the dominant period is estimated from the FFT peak of the context window and the average over all complete periods is tiled with phase alignment. Evaluated on ETTh1, ETTh2, and Weather at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

4.5.3 Observations

[OBS] Improved projection consistently collapses or reverses Panda’s advantage relative to vanilla on Weather. At $H = 96$: vanilla $+0.1592 \rightarrow$ improved -0.0380 . Chronos wins after improved decomposition. **[OBS]** At Weather $H = 336$, the improved projection reduces Panda’s advantage from $+0.1219$ to $+0.0299$ ($p = 0.012$, significant but small). **[OBS]** On ETTh1 and ETTh2, improved projection reduces or reverses Panda’s advantage at all tested horizons. **[OBS]** Panda’s absolute MAE increases after improved decomposition relative to vanilla on Weather. The improved projection does not reduce overall error; it changes the relative position of the two models.

[PAT] Across Experiments 11, 17, and 18 (Option A), three independent decomposition experiments consistently show that removing or isolating the periodic component collapses or reverses Panda’s advantage. This is the most reproducible finding of the mechanistic investigation.

Dataset	Condition	H	Panda advantage	p
Weather	Vanilla	96	+0.1592	0.008
Weather	Improved proj	96	−0.0380	1.000
Weather	Vanilla	336	+0.1219	0.008
Weather	Improved proj	336	+0.0299	0.012*
ETTh1	Vanilla	96	+0.0618	0.039*
ETTh1	Improved proj	96	+0.0551	0.191
ETTh1	Vanilla	336	+0.1005	0.125
ETTh1	Improved proj	336	−0.0382	0.320
ETTh2	Vanilla	96	−0.0015	0.371
ETTh2	Improved proj	96	−0.0546	0.875
ETTh2	Vanilla	336	−0.0671	0.371
ETTh2	Improved proj	336	−0.1426	0.961

Table 11: Experiment 17. Improved period projection versus vanilla. * denotes $p < 0.05$.

4.5.4 Competing Explanations

1. **[HYP]** Panda’s advantage on Weather requires the full mixed signal (trend, seasonal, and residual components jointly). Any decomposition disrupts the joint representation that Panda’s Koopman embedding exploits. **Confidence: Medium.:** consistent with all three experiments.
2. **[HYP]** Both Panda and Chronos are worse after decomposition because the improved projection still introduces errors, and Panda is more sensitive to input quality than Chronos. This makes Panda appear worse relative to Chronos, but the cause is input degradation rather than structural dependence on the full signal. **Confidence: Medium.:** the absolute MAE increase in both models is consistent with this.
3. **[HYP]** Chronos is specifically better than Panda at forecasting residuals. Once the periodic component is cleanly removed, Chronos recovers its advantage on the remaining stochastic component. **Confidence: Medium.:** consistent with the reversal at Weather $H = 96$.

Confidence: High. for the observational pattern. **Confidence: Low.** for any specific mechanism.

4.6 Experiment 18: Periodic Context with Real Targets

4.6.1 Motivation

Experiment 13 reported that Panda’s advantage amplifies on periodic-only context. However, the target in that experiment was a constructed FFT projection, not the real future values. Experiment 18 corrects this: both models are given the periodic-extracted context but evaluated against the actual future values of the Weather series. This directly determines whether the amplified advantage in Experiment 13 was real or a target-construction artifact.

4.6.2 Method

FFT extraction of top-5 frequency components from each context window channel (same as Experiment 13). Context presented to both models is the periodic component. Target is the real future values from the Weather series. Normalisation is applied using the statistics of the original (not periodic) context window, so the target normalisation is consistent with vanilla evaluation. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.6.3 Observations

Condition	H	Panda MAE	Chronos MAE	Advantage	p
Vanilla (reference)	96	0.6128	0.8021	+0.1893	0.004
Periodic ctx, real tgt	96	1.2007	1.0473	−0.1534	0.992
Vanilla (reference)	336	0.8754	0.9786	+0.1031	0.020
Periodic ctx, real tgt	336	1.1911	1.1816	−0.0095	0.875

Table 12: Experiment 18 (Option A). Both models evaluated against real future Weather values. Periodic context constructed from top-5 FFT harmonics of context window.

[OBS] With periodic context and real targets, Panda loses to Chronos at $H = 96$ (advantage -0.1534 , $p = 0.992$) and is essentially tied at $H = 336$ (advantage -0.0095 , $p = 0.875$). **[OBS]** Both models degrade substantially in absolute MAE relative to vanilla. Panda MAE increases from 0.6128 to 1.2007 (+96%). Chronos MAE increases from 0.8021 to 1.0473 (+31%). **[OBS]** Panda degrades approximately $3\times$ more than Chronos in absolute MAE when the context is reduced to its periodic component. **[OBS]** The amplified advantage in Experiment 13 (+0.52) is fully explained by the constructed target and does not reflect a real advantage of Panda on periodic signals.

4.6.4 Critical Finding

[PAT] The advantage Panda shows on Weather requires the full mixed signal. When given only the periodic component as context and evaluated against real targets, Panda loses to Chronos. This is confirmed across three independent experiments: Experiment 11 (decomp collapses advantage), Experiment 17 (improved decomp also collapses advantage), and Experiment 18 (periodic context with real targets reverses advantage).

4.6.5 Implication for TimesNet-Style Pretraining Direction

This result deprioritises TimesNet-style explicit period decomposition as an architectural modification. The evidence shows Panda performs *worse* on periodic signals in isolation, not better. Teaching Panda explicit period decomposition would address a component where Panda already underperforms Chronos, while potentially disrupting the full-signal joint representation that appears to be the source of its advantage.

4.6.6 Competing Explanations for Panda’s Larger Degradation

1. **[HYP]** Panda’s Koopman patch embedding is degenerate on smooth periodic inputs. Patches of a sinusoid look nearly identical after normalisation; the embedding cannot

distinguish position in the cycle. The forecast head defaults to a near-constant prediction. **Confidence: Medium.**: consistent with the scale of degradation.

2. **[HYP]** Chronos’s pretraining corpus is predominantly periodic and seasonal real-world data. Smooth sinusoidal inputs are near in-distribution for Chronos. **Confidence: Medium.**: consistent with Chronos degrading less.
3. **[HYP]** Panda is more sensitive to input amplitude calibration. The periodic context has lower energy than the full signal (residual energy removed), and Panda’s fixed prediction head scales poorly under this amplitude mismatch. **Confidence: Low.**: speculative, not directly tested.

Confidence: High. for the core observation. **Confidence: Low.** for any specific mechanism explaining Panda’s larger degradation.

5 Summary of Findings

5.1 Solid Observations (High Confidence)

1. Panda has statistically significant MAE advantage over Chronos on Weather at $H = 96, 192, 336$ ($n = 20, p \leq 0.001$). Advantage magnitude 0.17–0.24.
2. Channel attention does not drive this advantage. Univariate Panda performs comparably to or marginally better than multivariate Panda on Weather at $H = 96$ and $H = 336$ ($n = 8$, non-significant).
3. Panda’s Weather advantage requires the full mixed signal. Three independent experiments (Experiments 11, 17, 18) all show that removing or isolating the periodic component collapses or reverses Panda’s advantage. This is the most reproducible finding of the mechanistic investigation.
4. Panda wins on all tested chaotic `dysts` and Lorenz systems ($n = 8$ per system, no counter-example found).
5. The Lorenz ρ sweep shows a pattern consistent with a threshold near $\rho = 24.74$.

5.2 Medium Confidence Observations

1. Panda wins on Burgers at $\nu = 1.0$ (diffusion-dominated, non-chaotic by physical criteria), suggesting a chaos-specific mechanism does not fully explain the PDE advantage.
2. Panda’s advantage on Burgers increases monotonically with decreasing viscosity from $\nu = 1.0$ to $\nu = 0.01$, with a non-monotone reversal at $\nu = 0.005$.
3. Panda’s advantage on Lorenz $\rho = 28$ does not fully transfer to a phase-shuffled surrogate with matched power spectrum. The surrogate advantage (0.1715) loses significance at $n = 8$ windows.
4. Diversity subsampling does not consistently dominate stratified subsampling across 10 seeds. Single-seed subsampling conclusions are unreliable.
5. Panda absolute MAE is invariant to spatial subsampling method (Uniform, Stratified, Diversity) in Experiment 12. Whether this extends to heterogeneous real-world sensor data is not yet tested.

5.3 Not Established Despite Earlier Discussion

1. The mechanism behind Panda’s Weather advantage. Three experiments establish what it is not (channel attention, periodic handling in isolation) but the positive mechanism is not identified.
2. Whether chaos is necessary and sufficient for Panda’s advantage. Burgers result partially contradicts this for PDEs. Lorenz surrogate result is directionally consistent but underpowered.
3. Whether channel attention is the mechanism on any dataset.
4. Whether the Lorenz bifurcation result is chaos-specific or signal-statistics-specific. The surrogate experiment is consistent with chaos-specificity but underpowered.
5. Whether the Spearman correlation between λ_1 and Panda advantage is real (p-hacking issue from Experiment 4).
6. Whether λ_1 can be reliably estimated from Burgers PCA components. The Rosenstein estimator is inappropriate for this observable.

6 Proposed Next Experiments

6.1 Immediate: Sensor Heterogeneity Stratification on Weather

Scientific question: Is Panda’s absolute MAE on Weather limited by sensor heterogeneity? The Weather advantage requires the full signal (Experiments 11, 17, 18) and channel attention is not the driver (Experiment 9). A remaining hypothesis is that Panda’s Koopman embedding captures joint multi-channel structure but cannot adapt to heterogeneous sensor dynamics, since it was pretrained on homogeneous ODE state variables.

Proposed experiment: Compute dynamic feature vectors per Weather channel (standard deviation, lag-1 autocorrelation, spectral entropy, dominant frequency). Apply hierarchical clustering to group channels by dynamical similarity. Compare Panda absolute MAE on homogeneous subsets (same cluster) versus heterogeneous subsets (channels from different clusters). If Panda MAE is lower on homogeneous subsets, sensor heterogeneity is the bottleneck and G-SWaN-style node embeddings are the motivated architectural direction. If Panda MAE is invariant, consistent with Experiment 12, the bottleneck is elsewhere.

Falsification criterion: If Panda MAE coefficient of variation across heterogeneity levels is $< 5\%$ (consistent with Experiment 12 invariance), the heterogeneity hypothesis is not supported and the G-SWaN architectural direction should be reconsidered.

6.2 Architectural Direction: G-SWaN-Style Node Embeddings

Motivated by Prabowo et al. (G-SWaN, 2023), who observe that each real-world sensor has unique dynamics and that pairs of sensors also have unique pairwise dynamics. The paper proposes per-sensor node embeddings combined with a Spatial Graph Transformer (SGT) for adaptive pairwise attention.

This direction is motivated by two findings from this investigation: (1) Panda’s channel attention does not improve over univariate Panda on Weather, suggesting the current attention mechanism is not exploiting sensor identity; and (2) Panda absolute MAE is invariant to subsampling method on Burgers, suggesting the representation does not adapt to which sensors are provided.

Proposed experiment: Test Panda on a spatiotemporal dataset with measurable sensor heterogeneity (e.g. traffic sensor data). Compare Panda absolute MAE against a G-SWaN baseline on homogeneous versus heterogeneous sensor subsets. If Panda’s disadvantage relative to G-SWaN grows with heterogeneity, sensor identity is the missing component. If the gap is constant, heterogeneity is not the relevant factor.

6.3 Open Question: What Drives the Weather Advantage?

The positive mechanism behind Panda’s Weather advantage remains unidentified. Ruling out channel attention and periodic handling leaves the following candidates not yet tested:

1. **[HYP]** Panda’s Koopman patch embedding captures multi-scale temporal structure (joint trend, seasonal, residual) that Chronos’s causal univariate tokenisation cannot. This is specific to the joint representation and is destroyed by decomposition.

2. **[HYP]** Chronos is specifically poorly calibrated for Weather’s 21-channel multi-scale spectral structure (144-step daily cycle at 10-minute sampling). The advantage reflects Chronos weakness rather than Panda strength.
3. **[HYP]** Weather contains quasi-chaotic atmospheric dynamics at synoptic scales that activate Panda’s pretraining distribution.

Distinguishing hypothesis 1 from hypothesis 2 requires a Chronos-only ablation: measure Chronos MAE on Weather residuals (after decomposition) and compare to Chronos MAE on the full signal. If Chronos MAE is stable or improves on residuals, hypothesis 2 is supported (Chronos was specifically weak on the periodic component). If Chronos MAE also degrades after decomposition, hypothesis 1 is more likely.